5. Attempt any **two** parts of the following : **(10×2=20)**

(a) Explain the following with one example :

(i) Euler graph

(ii) Hamiltonian graph

(iii) Regular graph

(iv) Union of two graphs

(v) Isomorphic graphs.

(b) (i) Prove that a binary tree with n vertices has exactly (n + 1) null branches.

(ii) Let the pre-order and in-order search of a binary tree T yield the following sequence of vertices :

In-order	a	b	p	h	q	s	e	a	c	r	k	f	l

↑ root node

Pre-order	a	b	d	e	h	p	q	s	c	f	k	r	l

↑ root node

Draw the binary treetad also find the depth (hight) of T.

(c) Consider a DFA A = (Q, Σ, δ, q_1, F)

where, Q = { q_1, q_2, q_3, q_4} Σ = {0, 1} F = {q_4} and δ is given by the following table :

State/Σ	Input	
	0	1
→ q_1	q_2	q_3
q_2	q_3	q_1
q_3	q_1	q_4
(q_4)	q_2	q_2

Give the complete sequence of states for the input string 111000101 using the above transition table and find out whether string accepted by DFA 'A' or not.

(Following Paper ID and Roll No. to be filled in your Answer Book)

PAPER ID : 0934 Roll No.

B. Tech.

(SEM. IV) THEORY EXAMINATION 2010-11

DISCRETE MATHEMATICS

Time : 3 Hours *Total Marks : 100*

Note : Attempt **all** questions. Each question carries equal marks.

1. Answer any **four** parts of the following : **(5×4=20)**

(a) Let A, B, C be subsets of ∪. Given that A ∩ B = A ∩ C, is it necessary that B = C ? Justify your answer.

(b) Let A = {ϕ, b}, construct the following sets :

(i) A – {ϕ} (ii) {ϕ} – A

(iii) A ∪ P(A) (iv) A ∩ P(A)

(v) None of these

where P(A) is the power set of A.

(c) Let A = {1, 2, 3, 4} and

R = { (1, 2), (2, 3), (3, 4), (2, 1)}

Find the transitive closure of R.

(d) Define a relation matrix.

Let X = {1, 2, 3, 4} and R = {(x, y) : x > y}

Draw the graph of R and also give its matrix.

Source: AKTU official exam papers - www.aktupapers.com

(e) Let A = {1, 2, 3, 4}

Given an example of R in A which is :

(i) neither symmetric nor reflexive.

(ii) symmetric, reflexive but not transitive.

(iii) transitive and reflexive but not symmetric.

(f) Let $f : R \to R$ be defined by $f(x) = x^2$ and $g : R \to R$ be defined by $g(x) = x + 3$, then find out $(f \circ g) : R \to R$ and $(g \circ f) : R \to R$.

2. Attempt any **two** parts of the following : **(10×2=20)**

(a) Find out whether the following propositions are tautologies :

(i) $p \wedge (q \wedge r) \Leftrightarrow (p \wedge q) \wedge r$

(ii) $(p \wedge q) \Rightarrow (p \Rightarrow q)$

(b) Show that the following pair of propositions are logically equivalent :

$(p \vee q) \Rightarrow r$ and $(p \Rightarrow r) \wedge (q \Rightarrow r)$

Use truth table as well as algebra of proposition to show.

(c) Prove the validity of the following argument :

"If I get the job and work hard,

then I will get promoted.

If I get promoted, then I will be happy.

I will not be happy.

Therefore, either I will not get the job or

I will not work hard."

3. Attempt any **two** parts of the following : **(10×2=20)**

(a) Find the solution of following recurrence relation :

$a_r + 5a_{r-1} + 6a_{r-2} = 3r^2 - 2r + 1, r \geq 2$

given $a_0 = 2$ and $a_1 = 3$.

(b) (i) Find the generating function of a sequence $\{a_k\}$ if $a_k = 2 + 3k$.

(ii) Determine the discrete numeric function corresponding to the following generating function :

$$A(z) = \frac{1}{5 - 6z + z^2}.$$

(c) (i) A polygon has 44 diagonals. Find the number of its sides.

(ii) A committee of 5 is to be formed out of 6 males and 4 females. In how many ways this can be done when at least 2 females are included ?

4. Attempt any **four** parts of the following : **(5×4=20)**

(a) Let G be the set of all non-zero real numbers and let

$$a * b = \frac{ab}{2}$$

Show that $(G, *)$ is an abelien group.

(b) Determine whether a semigroup with more than one idempotent elements can be a group.

(c) Show that the system $(E, +, \cdot)$ of even integers is a ring under ordinary addition and multiplication.

(d) Show that the set $G = \{0, 1, 2, 3, 4\}$ is a finite abelian group of order 5 under addition module 5 as composition.

(e) Define subgroup. Let H be a subgroup of G. Then prove that :

(i) The identity of H is the same as that of G.

(ii) The inverse of any element of H is the same as the inverse of the same regarded as an element of G.

(f) Prove that the intersection of any two subgroups of a group G is again a subgroup of G.